Monoalphabetic Cipher

Mapping table:

- Plain Alphabet: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

- Cipher Alphabet: M K A Y Z R D B J C P E O T X F L W V G U S H Q N I

Encryption

1. Short Input "LOG"

follow the above mapping:

L → E

O → X

G → D

then ciphertext = EXD

2. Long Input " I LOVE DOING SOMETHING WHIMSY AND FUN"

follow mapping table:

I	→	J
LOVE	→	EXSZ
DOING	→	YXJTD
SOMETHING	→	VXO2GBJTD
WHIMSY	→	HBJOVN
AND	→	MTY
FUN	→	RUT

then ciphertext = J EXSZ YXJTD VXO2GBJTD HBJOVN MTY RUT

Decryption

+ Case 1: using mapping table

1. ciphertext = EXD

follow the mapping table:

E → L

X → O

D → G

therefore EXD → LOG

2. ciphertext = J EXSZ YXJTD VXO2GBJTD HBJOVN MTY RUT

follow the mapping table:

J	→	I
EXSZ	→	LOVE
YXJTD	→	DOING
VXO2GBJTD	→	SOMETHING
HBJOVN	→	WHIMSY
MTY	→	AND
RUT	→	FUN

therefore: J EXSZ YXJTD VXO2GBJTD HBJOVN MTY RUT → I LOVE DOING SOMETHING WHIMSY AND FUN

+ Case 2: without using mapping table

1. ciphertext: EXD

count frequency:

E = 1 (33.33%)

X = 1 (33.33%)

D = 1 (33.33%)

Since the letter have equal frequency, so it's impossible to analysis or identify the orginal word and there were many words that have three characters.

therefore, there many possible plaintext

LOG, CAT, DOG, PET, KID, BYE, LEG, ...

2. ciphertext: J EXSZ YXJTD VXOZGBJTD HBJOVN MTY RUT

by given ciphertext and count every letter by ignore space

J = 4 = $\frac{4}{31}$ x 100 = 12.903%

T = 4 = 12.903%

X = 3 = 9.677%

Z = 2 = 6.451%

Y = 2 = 6.451%

B = 2 = 6.451%

D = 2 = 6.451%

V = 2 = 6.451%

O = 2 = 6.451%

S = 1 = 3.225%

G = 1 = 3.225%

E = 1 = 3.225%

H = 1 = 3.225%

N = 1 = 3.225%

M = 1 = 3.225%

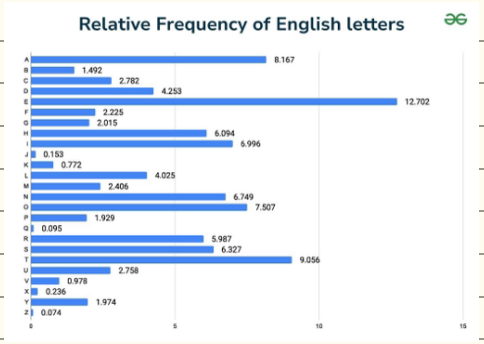
R = 1 = 3.225%

U = 1 = 3.225%

by total = 34 letters

follow by the formula

$$\text{percentage} = \frac{\text{letter_count}}{\text{total_letters}} \times 100\%$$



+ Step 1: guess using common English frequencies

So, we got the highest frequency letters in this ciphertext are J and T with 12.903%

We might initially guess that they represent E or T, since E and T are common in English with 12.702% and 9.056%, but these guesses aren't guaranteed.

+ Step 2: look the words patterns

ciphertext: J EXSZ YXJTD VXOZGBJTD HBJOVN MTY RUT

- J is a single letter word, so it is likely I or A
- YXJTD and VXOZGBJTD end in JTD, suppose to end with same plaintext
- MTY is a three letter word and might be AND or THE or another common word.

SO, we can see the sentence start with only letter and it should be subject.

then we can guess J → I

we have YXJTD and VXOZGBJTD but we known J → I, so these two word by ending with I__ then we can guess the verb word might ending with ING

I _____ __ING _____ING ___I___ ____

so, there will be J → I, T → N, D → G

I _____ __ING _____ING ___I___ __N__ ___N

after filled some letters we get one word in the middle of two words and contain N

_N_, so we can guess the word was AND ⇒ M → A, T → N, Y → D

I _____ __ING _____ING __I___ AND __N

now YXJTD becomes: D_ING, the likely word is DOING X→O

J EXS2 YXJTD VXO2GBJTD HBJOVN HTY AUT

I _D__ DDING _O_____ING __I___ AND __N

now we can see the word after subject, there will be verb. And this verb has 4 letters and contain _O__

so we can guess the common verb was LOVE

E→L, X→O, S→V, 2→E

I LOVE DDING _O_E__ING __I___ _N__ __N

on this moment we can get the word _O_E__ING, it likely SOMETHING

J EXS2 YXJTD VXO2GBJTD HBJOVN HTY AUT

I LOVE DDING SOMETHING __I___ AND __N

V→S, O→M, G→T, B→H

I LOVE DDING SOMETHING _HI___ AND __N

There fore, we guess the sentence: I LOVE DDING SOMETHING WHIMSY AND FUN

J→I E→L X→O S→V 2→E Y→D T→N D→G V→S O→M G→T B→H M→A R→F U→U